



Preparation and effect of vanadium addition on the mechanical properties of CoCrFeNiV_x high-entropy alloy

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ABSTRACT

High-entropy alloys (HEAs) have attracted considerable attention due to their unique phase structures and excellent properties. However, the existing preparation processes for HEAs involve high temperatures, lengthy procedures, and high costs, which limit their practical applications and confine them to the realm of fundamental research. Here, a novel method combining low-temperature solid-phase molten salt electrolysis and hot-press sintering was proposed to prepare CoCrFeNi HEA. Moreover, the study explored the addition of metal V, to modulate the composition of the HEA and improve its mechanical properties. The results showed that the molten salt electrolysis process could achieve the direct deoxidation of metal oxide mixtures, enabling one-step preparation of HEAs. During the early stages of deoxidation, the metals Ni and Co were obtained first and, under the influence of the mixed entropy, formed a non-stoichiometric NiCoCr alloy. As deoxidation progressed, Cr and Fe were successively reduced and further alloyed with the NiCoCr to form the non-stoichiometric CoCrFeNi HEA. Eventually, the aggregated dendritic metal Fe diffused and solid-solved into the non-stoichiometric CoCrFeNi HEA, resulting in the formation of a stoichiometric CoCrFeNi HEA. The CoCrFeNiV_x ($x = 0, 0.5, 1, 1.5$) alloy maintained a single-phase FCC structure, where V effectively improved the hardness of the CoCrFeNi HEA, with minimal impact on its phase structure. V played a role in solid-solution strengthening, grain refinement, and reducing Cr segregation in the alloy. Additionally, the appropriate addition of V simultaneously increased the tensile strength, hardness, and ductility of the CoCrFeNi HEA.

1. Introduction

As a multi-component alloy system, HEAs possess unique microstructures and structures due to their special multi-component solid solution effect. They exhibit excellent properties such as high strength, high toughness, high wear resistance, high temperature resistance, and oxidation resistance. HEAs have broad application potential in fields such as aerospace, ship transportation, nuclear processes, high-end manufacturing, electronics, and electrical engineering. The preparation process of HEAs is one of the limiting factors for their large-scale application. Developing a low-cost, low-energy, and short-process HEA preparation process is a prerequisite for the development of HEAs.

Electro-deoxidation process, proposed by Chen et al. [1], has been demonstrated that many metals and alloys could be obtained directly

from solid oxides in their pure and mixed forms. In the process, a pure and mixed forms metal oxide is used as the cathode, graphite is used as the anode, and molten CaCl₂ salt is used as the electrolyte. The potential applied in the cathode should polarize solid oxides to expel the oxide ions from the oxide into the molten salt sufficiently, but avoids the molten salt being electrolyzed. The oxide ions transport from solid oxides to the graphite anode in the electrolyte, and are discharged and then released by the form of carbon oxides [2]. The method has already been intensely studied in preparation of pure metals and alloys [3]. Therefore, it is reasonable to preparation HEAs in this method. The molten salt electrolysis process has advantages in preparing porous bulk or powder HEAs compared to other processes, such as low temperature, short process, and low-cost preparation. Numerous studies have demonstrated that this process can achieve the direct electrolytic deoxidation

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Table 1
Atomic ratio and composition ratio of CoCrFeNi HEA.

CoCrFeNi	Co/CoO	Cr/Cr ₂ O ₃	Fe/Fe ₂ O ₃	Ni/NiO	total
Atomic ratio (%)	25	25	25	25	100
Composition ratio (g)	0.1225	0.1250	0.13	0.1225	0.5

Table 2
Composition of the mixed oxide powders.

Number	x	CoO	Cr ₂ O ₃	Fe ₂ O ₃	NO	VO ₂	m total
V ¹ #	0	3.68	3.73	3.92	3.68	0	15
V ² #	0.5	3.24	3.28	3.45	3.24	1.79	15
V ³ #	1	2.89	2.93	3.08	2.89	3.20	15
V ⁴ #	1.5	2.61	2.65	2.79	2.61	4.34	15

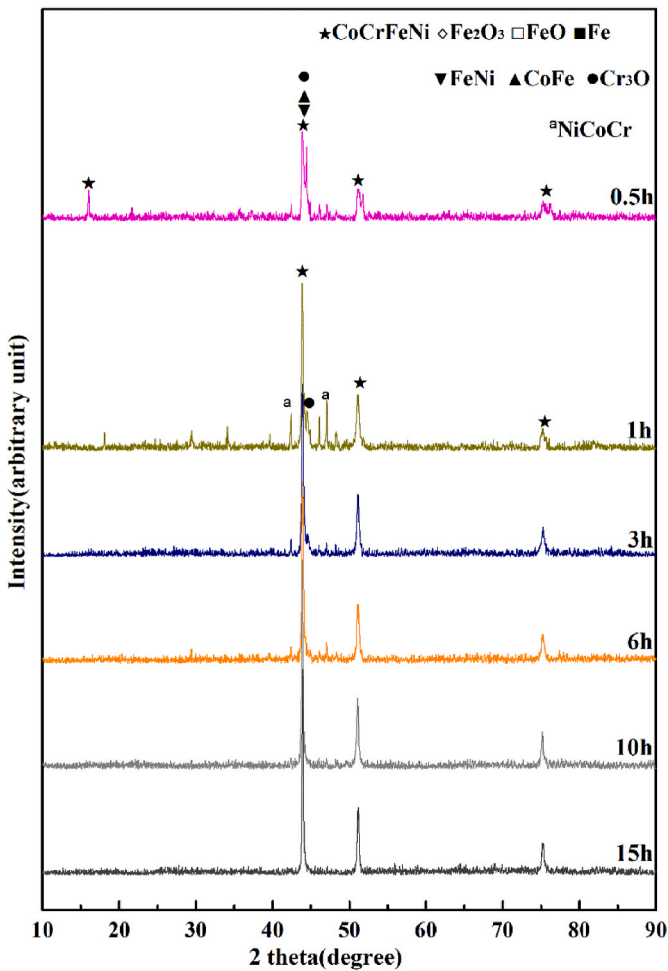


Fig. 1. XRD of the products with different electrolysis times.

preparation of metals or alloys from solid oxides [4–10]. However, research on the preparation of HEAs is still in the early exploration stage.

In 2017, Schwandt et al. successfully prepared equiatomic TiNbTaZr and TiNbTaZrHf refractory HEAs using molten salt electrolysis of mixed metal oxides [11]. The alloys exhibited uniform element distribution, and the bulk products showed a coupled nodular porous structure with high hardness. Subsequently, they successfully obtained HEAs in different forms, from bulk to powder, by adjusting the composition of the molten salt [12]. Wang et al. from Wuhan University successfully prepared FeCoNiCrMn HEA powder using this process [13]. The alloy powder was formed into a compact shape using spark plasma sintering,

and the corrosion resistance of the alloy was tested. The results showed that the HEA exhibited better corrosion resistance over a wide pH range compared to stainless steel. Jiao et al. from Beijing University of Science and Technology prepared AlCrNbTaTi HEA with a current efficiency of 46.7 % and identified research directions to improve the current efficiency [14].

Currently, the low current efficiency is the major challenge faced by the molten salt electrolysis process. Firstly, it is due to the harsh thermodynamic and kinetic conditions for the deoxidation of metal oxides. Secondly, it is caused by the parallel reaction of CO_3^{2-} at the cathodic electrochemical reduction, leading to carbon contamination in the products. However, Lv Zhaoping et al. discovered a new state of interstitial oxygen atoms in TiZrHfNb HEAs, called “ordered interstitial atom complexes” [15]. This structure significantly improves the strength and ductility of the alloy, demonstrating that an appropriate amount of oxygen can effectively regulate the trade-off between strength and ductility in HEAs. This research provides technical support for the design of high-strength and high-toughness metal materials. Furthermore, Baker et al. [16] studied the influence of carbon on the microstructure and mechanical properties of HEAs. The results showed that the stacking fault energy of HEAs decreases with increasing carbon content, leading to a transition from wavy slip to planar slip. Additionally, the formation of non-cellular structures induced by carbon greatly enhances the strain hardening and ultimate tensile strength of HEAs. The delayed necking caused by high strain hardening, as well as plastic accommodation through microbands and regional boundaries, increases the ductility of HEAs with the addition of carbon. This research breaks the conventional understanding that carbon content and ductility are contradictory in traditional alloys, suggesting that the ductility of HEAs can also be adjusted by adding a certain amount of carbon. This provides theoretical support for the preparation of high-performance HEAs with excellent mechanical properties through molten salt electrolysis.

In summary, the molten salt electrolysis process can achieve the direct deoxidation of solid oxides for the preparation of HEAs. By controlling the oxygen or carbon content in HEAs, it is possible to improve their strength and ductility. This provides favorable conditions for the efficient preparation of high-performance HEAs using molten salt electrolysis. This thesis adopts the molten salt electrolysis process to prepare CoCrFeNi HEAs and clarifies the deoxidation and alloying mechanisms of solid oxide mixtures. The mechanical properties of CoCrFeNi HEAs are improved by adding metals V, and its effects on the properties of HEAs are analyzed.

2. Experimental

The molten salt deoxidation experiment is conducted in a vertical resistance furnace at a temperature of 1173 K. The anode is a grinding rod, and the cathode is a mixture of metal oxides prepared according to the equiatomic ratio shown in Table 1. The mixture powder is pressed

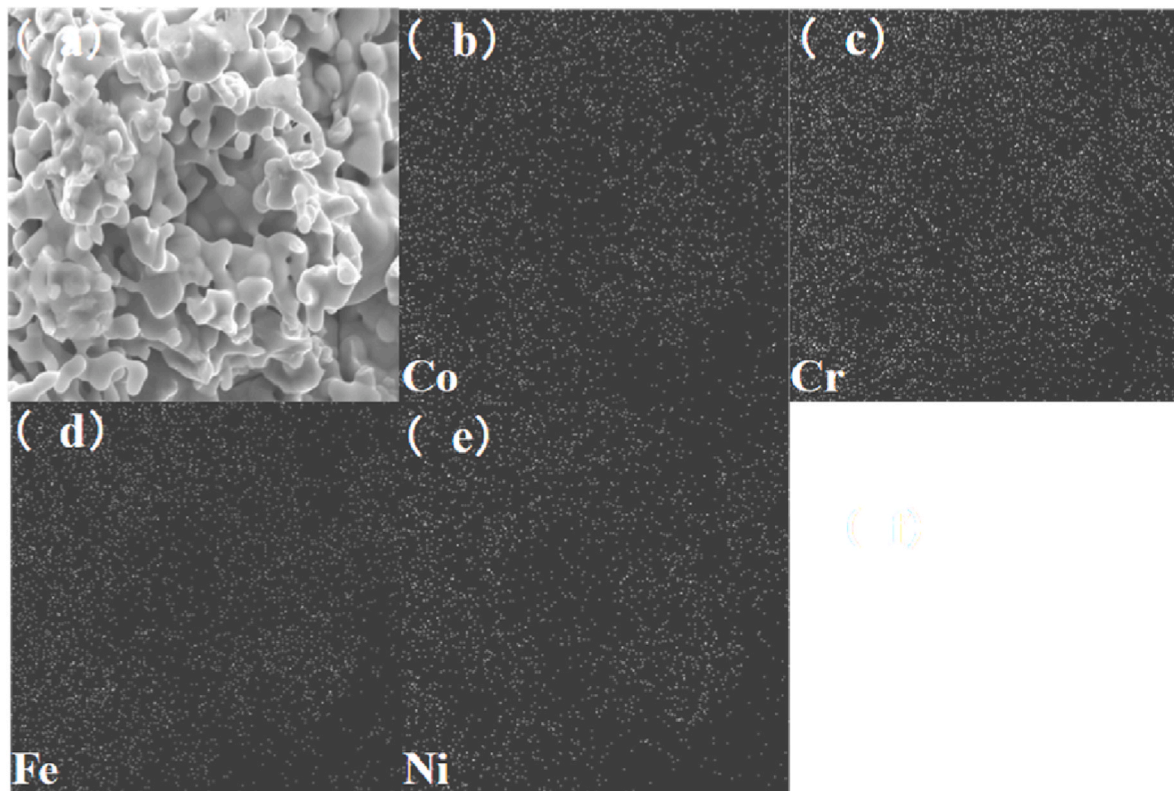


Fig. 2. Morphology and surface scanning results after electrolysis.

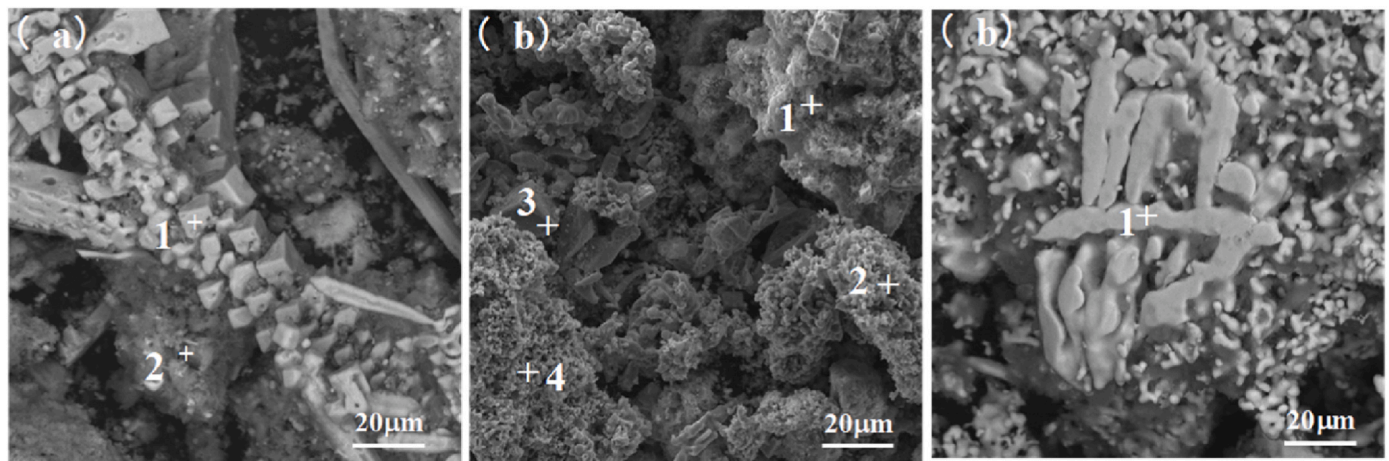


Fig. 3. BSE images and EDX analysis of the product with 15 min(a), 30min(b), 1 h(c) electrolysis time.

into a block using a cylindrical mold with a diameter of 20 mm at a pressure of 4 MPa. Then, the block is dried at a temperature of 353 K for 1 h. Subsequently, electrolysis is performed in a CaCl_2 molten salt at 1173 K to obtain CoCrFeNi HEA.

Table 2 shows CoCrFeNi $_x$ (V#) HEAs with different contents of V. The HEA powders were subjected to hot pressing at 1373 K and a pressure of 30 MPa for 1 h to obtain bulk HEAs.

The crystal structure of the raw materials and HEA products was determined using a D/max 2500 PC X-ray diffractometer produced by Rigaku Corporation in Japan. The sample morphology and microstructure were observed using scanning electron microscopy (SEM) with a TESCAN VEGA II equipped with an Oxford INCA Energy 350 tungsten filament and a Thermo Fisher Scientific Quattro S FESEM. Both SEMs were equipped with energy-dispersive X-ray spectroscopy (EDX).

The room temperature strength of the HEA was tested using a WDW-100 electronic universal material testing machine. The tensile strain rate was set to 1×10^{-4} /s. Prior to the tensile tests, the dog bone-shaped samples were polished using sandpaper from 600# to 3000# to ensure there were no visible scratches on the sample surface. Each alloy composition was tested three times to ensure the accuracy of the results. The hardness of the alloy was determined using a Model HV-115 Vickers hardness tester produced by Kanagawa Corporation in Japan. The samples were first polished with sandpaper up to 3000#, followed by polishing. The applied load was determined based on the sample hardness, with a load holding time of 10 s. The hardness data were obtained by measuring the hardness at 10 different positions on the sample surface and calculating the average value.

Table 3

EDX results of electro deoxidation products for 15min, 30min and 1 h

Time	point	Element	Wt.%	At.%
15min	1	Cr K	3.98	4.29
		Fe K	84.33	84.60
		Co K	11.69	11.11
	2	O K	28.79	56.30
		Ca K	12.55	9.79
		Cr K	34.90	21.00
		Fe K	9.78	5.48
30min	1	Co K	3.99	7.43
		O K	24.92	37.58
		Ca K	8.8	5.97
		Cr K	19.23	17.83
		Fe K	16.58	13.13
		Co K	17.47	13.43
	2	Ni K	15.31	12.06
		O K	2.25	7.61
		Cr K	16.51	17.2
		Fe K	6.13	5.95
		Co K	27.31	25.11
		Ni K	47.81	44.13
	3	Cr K	5.94	6.36
		Fe K	94.06	93.64
	4	Cr K	20.28	22.03
		Fe K	28.78	29.11
		Co K	35.58	34.1
		Ni K	15.35	14.77
1 h	1	Cr K	18.64	19.96
		Fe K	59.56	59.38
		Co K	6.57	6.21
		Ni K	15.23	14.45

3. Results and discussion

3.1. Preparation and alloying mechanisms of CoCrFeNi HEAs

The precursor deoxygenation process was studied by selecting products after deoxygenation for 15 min, 0.5 h, 1 h, 3 h, 6 h, 10 h, and 15 h. Fig. 1 shows the XRD patterns of the products at different deoxygenation times. The change in diffraction peaks with deoxygenation time indicates that the precursor completely transforms into CoCrFeNi HEA after 3 h of electrolytic deoxygenation. No phase change occurred with longer deoxygenation times. The XRD pattern of the product at 5 min of deoxygenation indicates the presence of metallic Ni, Co, and CaCr_2O_4 . The formation of Ni and Co can be attributed to the large initial reaction interface between the cathodic oxide spheres and the stainless-steel mesh, as well as their good conductivity. Additionally, the low theoretical decomposition potential and metal valence states make NiO and CoO more easily reduced to pure metals. CaCr_2O_4 formation was also observed in the reduced product. Derek J. Fray [17,18] conducted a detailed study on this phenomenon, and the results showed that calcium ions temporarily exist in the cathode and play a positive role in the reduction process.

Fig. 2 shows the morphology and surface scanning results after electrolysis, indicating a uniform distribution of elements in the electrolytically obtained HEAs. Fig. 3 presents the backscattered electron (BSE) images and EDX analysis of the products obtained after 15 min, 30 min, and 1 h of deoxygenation. Table 3 provides the EDX results for the products obtained after 15 min, 30 min, and 1 h of electrolytic deoxygenation. Fig. 3(a) shows the BSE image of the product obtained after 15 min of deoxygenation. The EDX results indicate that the dendritic structure at Point 1 is composed of Fe, and the morphological differences may be attributed to the varying contents of Cr and Co. The analysis at Point 2 reveals that the matrix phase is mainly composed of CaCr_2O_4 and oxides. The XRD results of the product obtained after 0.5 h of deoxygenation indicate the disappearance of the spinel phase in the precursor material after 0.5 h. The un-reduced oxygen-containing phases in the cathode product are CaCr_2O_4 and Cr_3O . In addition to the main phase of

CoCrFeNi HEA, the cathode product also contains metallic Fe phase. The EDX results at Point 2 indicate that the nodular structure corresponds to a non-stoichiometric CoCrFeNi HEA. The phase at Point 1 consists of un-reduced CaCr_2O_4 and Fe, which did not alloy with other elements to form a HEA. According to the XRD results, the product obtained after 1 h of deoxygenation consists of CoCrFeNi HEA, metallic Fe, σ phase, and Cr_3O . Iron, after being reduced to metallic Fe during the molten salt deoxygenation process, aggregates into dendritic structures. Subsequently, with prolonged deoxygenation, metallic Fe diffuses and forms a solid solution with other metallic elements, resulting in the CoCrFeNi HEA phase.

After 3 h of electrolytic deoxygenation, the product consists only of the CoCrFeNi HEA phase. Fig. 4(a–d) show the BSE images and EDX analysis of the products obtained after deoxygenation for 3–15 h. The structure of the CoCrFeNi HEA appears as nodular porous structures that are interconnected along the surface of the particles. The deoxygenation time does not have a significant influence on the particle size and structure of the HEA. The oxygen content, current efficiency, and energy consumption of the CoCrFeNi HEA after the whole fabrication process were measured and calculated, which was published in the literature [19].

Fig. 5 depicts the deoxidation process of CoCrFeNi HEA prepared by molten salt electro-deoxidation. As shown in the diagram, the deoxidation process of CoCrFeNi HEA includes electrochemical reduction and alloying.

During the electrochemical reduction process, NiO and CoO, due to their low decomposition voltage and oxide valence state, are easily reduced to elemental metals. Cr_2O_3 reacts with Ca^{2+} under the influence of electric current to form CaCr_2O_4 and release a small amount of elemental Cr. Subsequently, CaCr_2O_4 is electrolyzed to generate metallic Cr. The deoxidation of Fe_2O_3 involves a process of $\text{Fe}_2\text{O}_3 \rightarrow \text{FeO} \rightarrow \text{Fe}$, leading to the formation of metallic Fe that aggregates into dendritic structures. The alloying process begins with the formation of metallic phases. In the early stage of deoxidation, the first metals to be reduced, Ni and Co, along with the small amount of released Cr, form a non-stoichiometric NiCoCr alloy under the influence of mixing entropy. This stage corresponds to the deoxidation time of 5 min. As deoxidation progresses, Cr reacts with Fe, and under the influence of mixing entropy, further alloying occurs with the NiCoCr alloy, resulting in the formation of a non-stoichiometric CoCrFeNi HEA. This stage corresponds to the deoxidation time of 15 min. Finally, the aggregated dendritic metallic Fe diffuses and solid-solves with the non-stoichiometric CoCrFeNi HEA, ultimately forming a stoichiometric CoCrFeNi HEA.

3.2. Effect of V on the performance of CoCrFeNi HEA

Fig. 6 shows the XRD spectra of CoCrFeNiV_x alloys with different V contents. From the spectra, it can be observed that all CoCrFeNiV_x alloys with different V contents are composed of a single-phase FCC solid solution. Research has indicated that when $x > 0.7$ in CoCrFeNiV_x HEAs, a tetragonal σ phase with a square crystal structure exists in the alloy, and the peak intensity of this phase increases with the increase in V content. Senkov et al. [20] proposed the phase transformation behavior of CoCrFeNiV HEAs, suggesting that the first phase to solidify is a high-temperature phase, which serves as a precursor to a coexisting microstructure like a eutectoid structure. This phase may have a BCC structure, similar to high-temperature δ -ferrite in steels. The second phase to solidify is an FCC phase at grain boundaries. Upon cooling, the high-temperature BCC phase undergoes a eutectoid transformation into a mixture of the σ phase and FCC phase. In this study, the XRD spectra did not show peaks corresponding to the σ phase, which is attributed to the fact that the temperature during the hot-press sintering process was significantly lower than the melting point of the metals, and the alloy did not undergo liquid phase solidification. Therefore, the high-temperature BCC phase and the subsequent eutectoid transformation into a mixture of the σ phase and FCC phase could not occur.

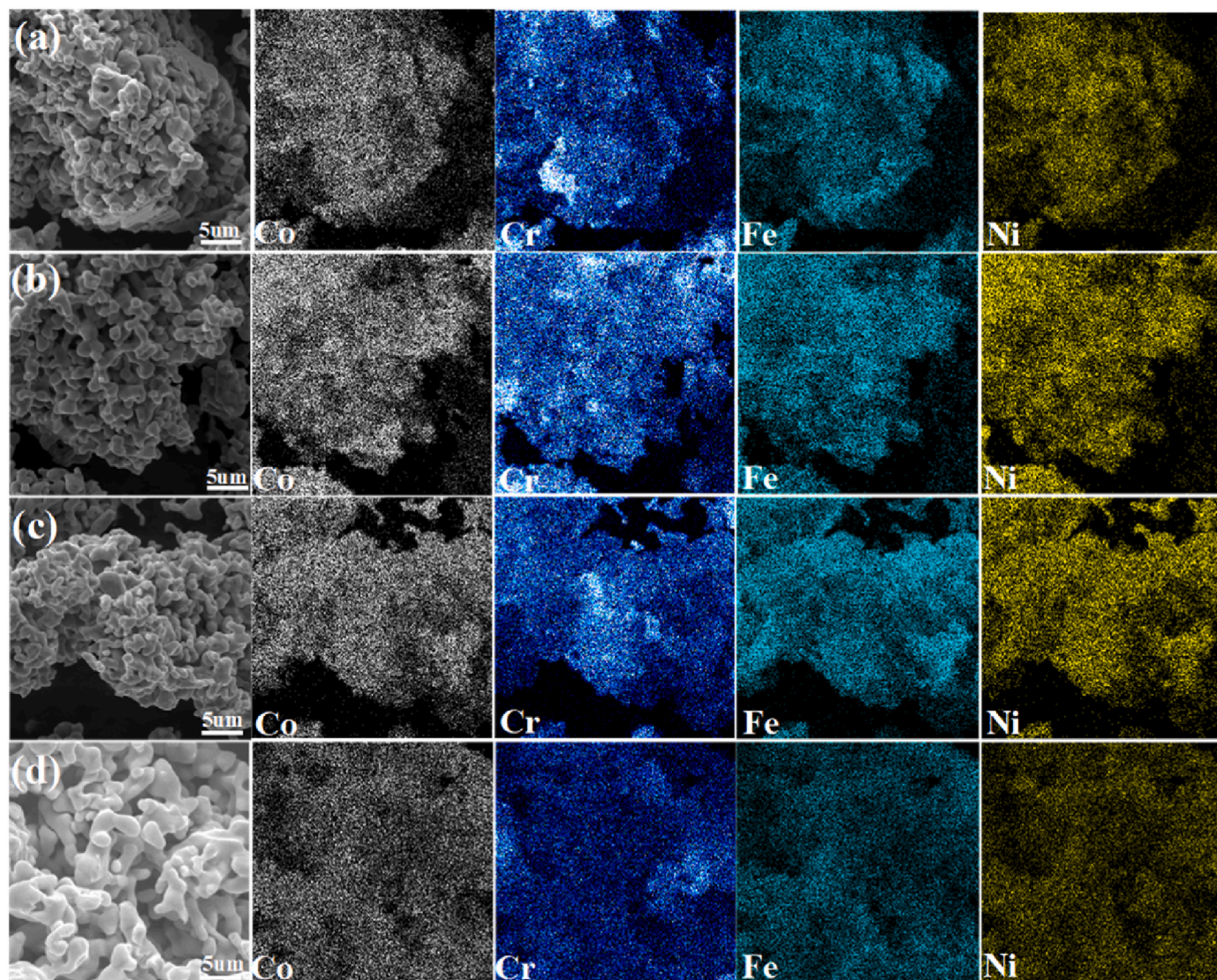


Fig. 4. BSE images and EDX analysis of the product from 3 h to 15 (a–d) h electrolysis time.

during the hot-press sintering process. The relative density for the bulk CoCrFeNiTi_x HEAs with different V contents was listed in Table 4.

Fig. 7 shows the BSE images and corresponding EDX analysis of CoCrFeNiV_x alloys with different V contents. From the low-magnification BSE images, it can be observed that dark-colored phases are dispersed within the HEA matrix, indicating the presence of segregation and clustering of certain elements or phases in the sample. From the high-magnification BSE images, it can be noted that the size of the dark-colored phases in region B initially decreases (V0–V1.5) with increasing V content. The EDX analysis reveals that the dark-colored phases in the V0 alloy of region B are segregated Cr and C elements. With the addition of V, V reacts with C to form more chemically stable VC, which significantly reduces the segregation of Cr. In contrast to the nearly continuous reticular σ phase formed in the as-solidified state of CoCrFeNiV and CoCrFeNiMnV HEAs prepared by Senkov et al. through arc melting, the CoCrFeNiV_x alloys ($x = 0.5, 1.0, 1.5$) in this study exhibit fine irregularly shaped particles with Cr and V segregation, along with the presence of C elements, distributed uniformly throughout the FCC phase matrix, even at higher observation magnifications. However, when the V content reaches V1.5, the distribution of Cr becomes more uniform, but the aggregation size of V increases, possibly due to the poor

compatibility of V with other alloying elements, leading to significant V clustering in the alloy [20].

Fig. 8 shows the tensile stress-strain curves and fracture surface morphology of CoCrFeNiV_x alloys. With the addition of V, the tensile strength and ductility of the CoCrFeNiV_x alloys initially increase and then decrease. The HEAs of V0, V0.5, and V1.0 exhibit a gradual increase in tensile strength. It can be observed that the addition of V promotes solid solution strengthening in the alloy. The larger atomic radius of vanadium causes lattice distortion when it is dissolved in the matrix, increasing the resistance to dislocation motion and resulting in solid solution strengthening. Therefore, the addition of vanadium enhances the hardness of the alloy. Additionally, metallic V has the effect of refining the grain size, leading to grain refinement strengthening. Moreover, the high binding energy between V and carbon allows the formation of fine VC dispersed particles, which reduces the segregation of Cr and contributes to the strengthening effect of second-phase particles. The increase in ductility of the V0, V0.5, and V1.0 HEAs is mainly attributed to the grain refinement caused by the addition of V and the improved uniformity of alloy element distribution, reducing the segregation of Cr. However, when the V content reaches V1.5, both the tensile strength and ductility of the alloy significantly deteriorate due to the

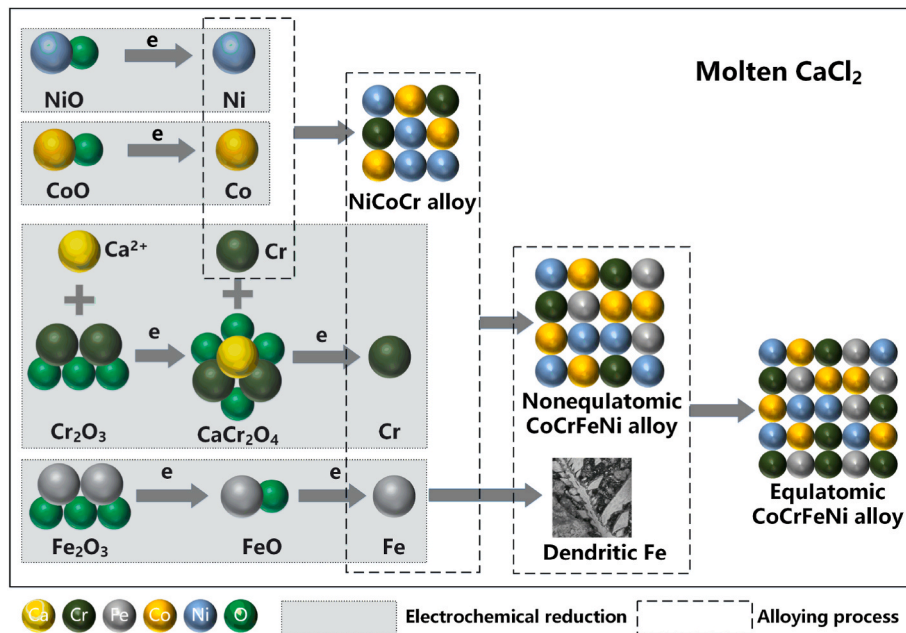


Fig. 5. Schematic diagram of deoxidation process of CoCrFeNi HEA.

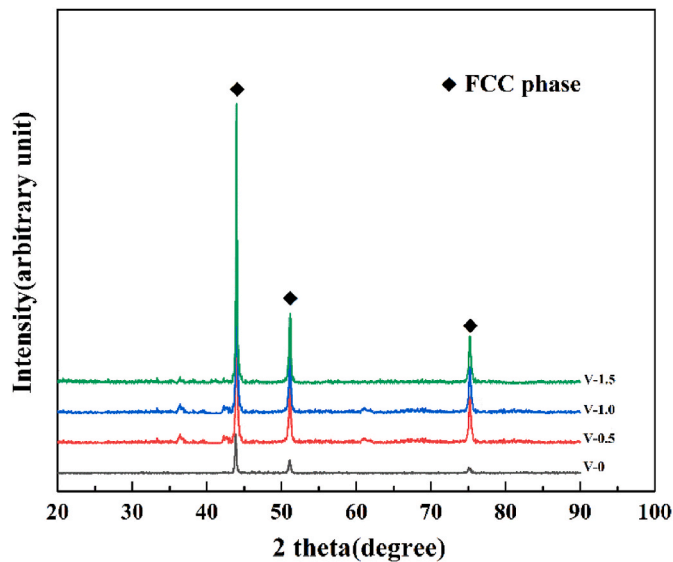


Fig. 6. XRD patterns of the bulk CoCrFeNiV_x HEAs with different V contents.

Table 4

Relative density of the bulk CoCrFeNiV_x HEAs with different V contents.

Samples	V0	V0.5	V1.0	V1.5
relative density (g/cm ³)	8.17	7.94	7.76	7.61

pronounced segregation and agglomeration of V, as well as the higher oxygen content in the alloy.

From Fig. 8, it can be observed that the fracture surfaces of CoCrFeNiV_x HEAs exhibit many dimples, indicating typical ductile fracture behavior. Dimples are characteristic features left on the fracture surface of alloy materials, formed by the nucleation, growth, and aggregation of microcracks during plastic deformation, and reflect the energy absorption capacity of the material. From panels (a) to (c), it can be seen that

the size of the dimples gradually decreases and becomes more uniform with increasing V content, which is attributed to the grain refinement caused by the addition of V. The smaller and more uniform dimples further confirm the increase in alloy strength and ductility. Panel (c) shows the fracture morphology of the V1.0 alloy, where numerous fine and deep dimple structures are distributed on the fracture surface, along with some small particles at the bottom of the dimples. These small particles may act as preferential nucleation sites for micro voids or microcracks during the tensile process. With the increasing tensile strain, these micro voids or microcracks coalesce, leading to the formation of localized large cracks, ultimately resulting in fracture failure of the composite material. Panel (d) presents the fracture morphology of the V1.5 alloy, where a large number of dimples are still present, exhibiting typical ductile fracture characteristics. However, compared to alloys with lower V content, there are more pronounced particles and cracks on the fracture surface of the V1.5 alloy, which can be attributed to the segregation and agglomeration of V as well as the higher oxygen content in the alloy.

Fig. 9 shows the hardness of CoCrFeNiV_x alloys with different V contents. It can be observed that the addition of V significantly enhances the hardness of CoCrFeNiV_x alloys, with a hardness value of 451 H V for the V1.5 alloy.

4. Conclusions

CoCrFeNi HEA can be obtained by electro-deoxidation process in molten salt at low temperature and short time. During the deoxidation process, the first metals to be reduced are Ni and Co, which, in the presence of a small amount of Cr, form a non-stoichiometric NiCoCr alloy due to the mixing entropy. As deoxidation progresses, Cr and Fe are continuously reduced and further alloy with the NiCoCr alloy, forming a non-stoichiometric CoCrFeNi HEA. Finally, dendritic metallic Fe diffuses and solidifies with the non-stoichiometric CoCrFeNi HEA, resulting in a stoichiometric CoCrFeNi HEA.

Vanadium addition can effectively improve the hardness of CoCrFeNi HEA without altering their phase structure. V contributes to solid solution strengthening, grain refinement, and reducing the segregation of Cr in the alloy. The addition of V significantly increases the hardness of the alloy. An appropriate amount of V addition can simultaneously increase the tensile strength and hardness of CoCrFeNi HEA. The V1.5

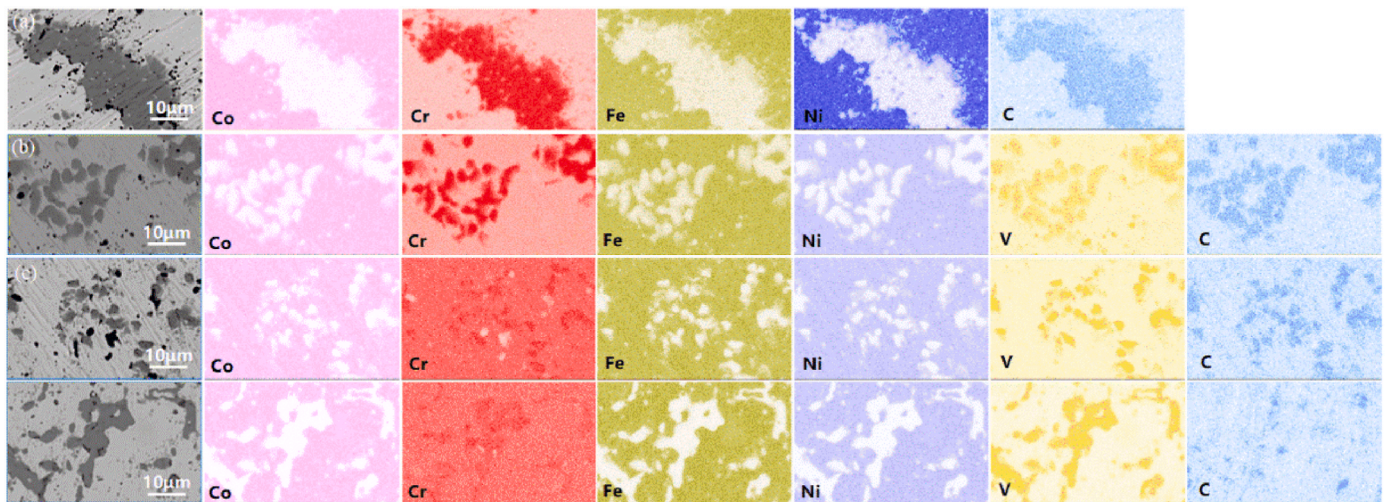


Fig. 7. BSE images and EDX analysis of the samples, (a) V_0 , (b) $V_{0.5}$, (c) $V_{1.0}$, (d) $V_{1.5}$.

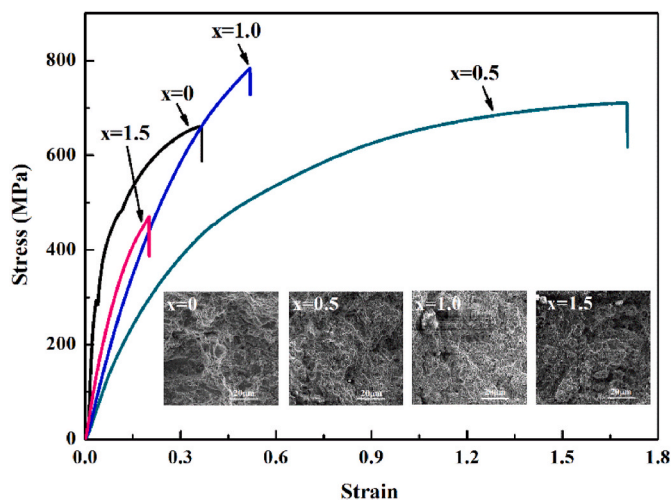


Fig. 8. Stress-strain curves of CoCrFeNiV_x HEAs and Fracture surfaces of CoCrFeNiV_x HEAs.

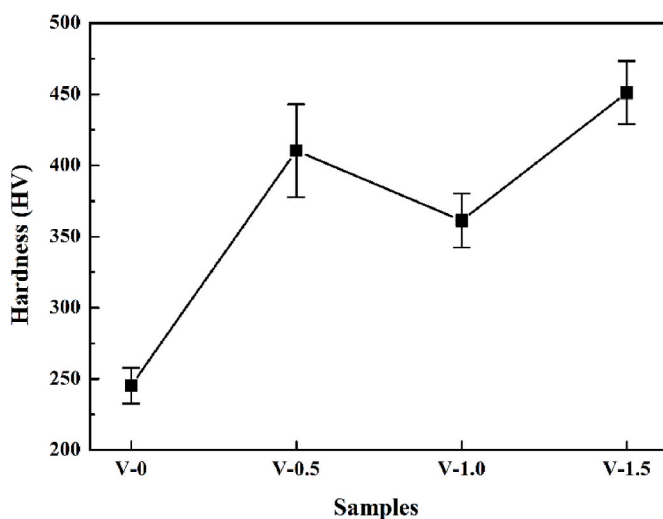


Fig. 9. Hardness of CoCrFeNiV_x HEAs and ultimate tensile strength curve.

alloy has a hardness of 451 H V, which is higher than that of the alloy without V addition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Mengjun Hu reports financial support was provided by Chongqing Natural Science Foundation (cstc2021jcyj-msxmX1049).

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